



# Mario-Cristian Constantinescu

📍 **Home** : Romania

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🌐 **Website**: <https://mario-cristian.github.io/portfolio/>

**Gender**: Male **Date of birth**: 06/05/2002 **Nationality**: Romanian

## EDUCATION AND TRAINING

[ 10/2024 – Current ]

### Master's student

*Universitatea Ovidius, Facultatea Matematica si Informatica*

**City**: Constanta | **Country**: Romania | **Field(s) of study**: Cyber Security and Machine Learning

[ 11/2024 ]

### Unihack Hackathon Timișoara (48h)

**City**: Timișoara | **Country**: Romania |

**Link**: <https://mario-cristian.github.io/portfolio/>

We participated as a team of 4 on the education track: Empowering Tomorrow's Minds with Innovative Education.

Our goal is to demonstrate that math can solve real-world problems, not just theoretical ones, and to show how math is used in machine learning using AI tools.

Our solution revolves around live, interactive experiments with a gamification element.

- Teachers can create lessons (AI-generated with optional edits) and attach experiments to them.
- Students can engage with interactive experiments that help them visualize concepts and solve problems.

Technologies used:

- Streamlit
- Flask
- MongoDB

[ 2021 – 2024 ]

### Graduated in Computer Science

*Universitatea Ovidius, Facultatea Matematica si Informatica*

**City**: Constanta | **Country**: Romania | **Final grade**: 9.87

During 3 years at the university I built several projects and I learnt the following technologies/programming languages:

- C/C++
- Data structures
- Java (OOP, Swing, AWT)
- Python (Numerical methods)
- Machine learning
  - Linear Regression
  - Logistic Regression
  - K-Nearest Neighbor
  - Naive Bayes
  - Decision Tree
  - Random Forest
  - SVM
  - K-Means
  - Artificial Neural Network
  - Convolutional Neural Network
  - TensorFlow

- Web Development
  - Frontend (HTML, CSS, JavaScript, jQuery, AJAX)
  - Backend (ASP.NET, Django)
- Docker
- Databases
  - Oracle
  - SQL Server
  - SQL, PL/SQL
- Web graphic (Three.js)

[ 05/2024 ] **Estic 2024**

**City:** Constanta | **Country:** Romania |

I presented my final project: web application for students and teachers with physics simulations that can be controlled through gestures recognized by the camera.

For the frontend I used:

- HTML
- CSS
- JavaScript
- Bootstrap
- jQuery
- AJAX
- Web Sockets

For the backend I used:

- Django
- MediaPipe
- TensorFlow
- OpenCV
- Web Sockets

[ 04/2024 ] **iTEC Hackathon Timișoara (48h)**

**City:** Timișoara | **Country:** Romania |

- participated as a team of 3 members and we developed a web application that monitorizes the endpoints of other applications
- I was part of both the frontend and backend teams
- for the frontend we used HTML, CSS, jQuery, Bootstrap, AJAX
- for the backend we used ASP.NET, Dapper, stored procedures (PL/SQL)

[ 03/2024 ] **IT Fest Hackathon Timișoara (48h)**

**City:** Timișoara | **Country:** Romania |

- we participated as a team of 4 people and we developed a web application that had the aim to encourage a healthy lifestyle through workouts and nutrition
- I was part of the frontend team
- for the frontend we used Angular
- for the backend we used ASP.NET, Dapper, stored procedures (PL/SQL)

[ 12/2023 ] **BEST Brasov Hackathon (24h)**

**BEST Brasov**

**City:** Brasov | **Country:** Romania |

- we participated as a team of 3 members and we developed a web application where teachers could post their own courses and the students could access and learn from them
- I was part of the frontend team
- for the frontend we used Angular
- for the backend we used ASP.NET, Dapper, stored procedures (PL/SQL)

[ 10/2023 ] **Axes Hackathon (24h)**

## Axes

**City:** Bucuresti | **Country:** Romania |

- we participated as a team of 4 people and we developed a web application for monitoring the electricity consumption and CO2 footprint of a person
- I was part of the frontend team
- for the frontend we used Angular
- for the backend we used ASP.NET, Dapper, stored procedures (PL/SQL)

[ 12/2022 – 07/2023 ] **Angular and C# ASP.NET practice**

### *Corner Stone Technologies*

**City:** Constanta | **Country:** Romania |

[ 05/2023 ] **Estic 2023**

**City:** Constanta | **Country:** Romania |

I developed a web application for dating and I used the following technologies:

- for the frontend: Angular
- for the backend: ASP.NET

[ 07/2022 ] **Microsoft Azure practice**

[ 09/2017 – 06/2021 ] **High school graduate**

### *Liceul Teoretic "Carmen Sylva"*

**City:** Eforie Sud | **Country:** Romania | | **Final grade:** 9.01

- Participation in math and informatics olympiads in the 9th and 10th grade (first phase)

## PROJECTS

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### Generative AI / Natural Language Processing (NLP)

- Large Language Model (LLM) from scratch using character tokenization

**Link:** <https://mario-cristian.github.io/portfolio/>

### Machine Learning (ML)

- Neural Network from scratch
- Gaussian Mixture Model (GMM) with Expectation-Maximization (EM) from scratch
- Linear Regression from scratch
- Logistic Regression from scratch
- Gaussian Naive Bayes from scratch
- Decision Tree from scratch
- Random Forest from scratch
- K-Nearest Neighbor with PCA - face recognition from scratch
- K-Means from scratch
- Search Engine from scratch

**Link:** <https://mario-cristian.github.io/portfolio/>

### Geometry

Made with Java AWT.

- Point inside or outside a simple polygon
- Point inside or outside a convex polygon
- Point inside or outside PSLG
- Points inside a rectangle

**Link:** <https://mario-cristian.github.io/portfolio/>

## LANGUAGE SKILLS

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**Mother tongue(s):** Romanian

**Other language(s):**

**English**

**LISTENING** B2 **READING** B2 **WRITING** B2

**SPOKEN PRODUCTION** B2 **SPOKEN INTERACTION** B2

**Spanish**

**LISTENING** B1 **READING** B1 **WRITING** B1

**SPOKEN PRODUCTION** B1 **SPOKEN INTERACTION** B1

*Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user*

## DRIVING LICENCE

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**Cars:** B